

SPECTRA-HET SPLIT UP: EXACT U -STATISTIC HETEROSKEDASTIC LINEAR-SHRINKAGE ADAPTIVE GMM FOR UNPAIRED CAUSAL INFERENCE

Anonymous authors

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ABSTRACT

Estimating structural causal effects from *unpaired* multi-environment data under hidden confounding is a fundamental challenge across biomedical research, econometrics, and the physical sciences. Existing two-sample instrumental-variable (TS-IV) methods suffer from four interacting failure modes: high-dimensional-instrument measurement-error bias, random sample-splitting partition variance, efficiency loss under environmental outcome-noise heteroskedasticity, and spectral noise inflation or false-positive coefficient blow-up in high feature dimensions. We introduce **SPECTRA-HET SPLIT UP**, a unified Generalized Method of Moments (GMM) framework that resolves all four within a single estimator through: (1) an *exact* closed-form Leave-One-Out (LOO) U -statistic cross-moment operator that attains zero split variance and equals the infinite split-averaging limit ($H \rightarrow \infty$) in one deterministic pass; (2) feasible generalized-least-squares (FGLS) outcome-noise reweighting across environments; (3) Ledoit–Wolf linear-shrinkage precision weighting that keeps the moment matrix well-conditioned across arbitrary aspect ratios d/m ; and (4) an adaptive score-penalized Lasso ($\gamma = 1.0$) with guarded post-selection refitting ($\tau = 0.1 \times \max_j |\hat{\beta}_j|$) that removes attenuation bias while suppressing false positives. Across synthetic high-dimensional regimes, real physical experiments on the Causal Chambers light tunnel and wind tunnel, and genome-scale single-cell Perturb-seq data (CausalBench, $d=100$), SPECTRA-HET SPLIT UP reduces L_1 estimation error by up to $8\times$ relative to standard SPLIT UP, heteroskedasticity-robust econometric estimators (HLIM/HFUL, anchor regression), and other TS-IV baselines. Ablations isolate each component, and an environment-block bootstrap delivers valid post-selection confidence intervals.

1 INTRODUCTION

Estimating structural causal effects under hidden confounding is a fundamental challenge across biomedical research, econometrics, the physical sciences, and robotics (Yao et al., 2020; Fang & Liang, 2024). In many realistic settings, treatment covariates $X \in \mathbb{R}^d$ and outcomes $Y \in \mathbb{R}$ cannot be recorded in a single joint sample. Instead, practitioners observe *unpaired* data collected across multiple environments $I \in \{1, \dots, m\}$ —for instance single-cell perturbation benchmarks (Chevalley et al., 2022), robotic manipulation under physical interventions (Ahmed et al., 2020), multi-environment corpora (Sobhani et al., 2024), or separate epidemiological cohorts (Bowden et al., 2015). When environment indicators act as categorical instruments, causal-effect estimation becomes a two-sample instrumental-variable (TS-IV) problem under exclusion restrictions and cross-environment moment invariance.

Despite theoretical identifiability, existing TS-IV and multi-environment estimators exhibit four severe, interconnected failure modes on finite unpaired samples:

1. **High-dimensional-instrument measurement-error bias.** When the number of environments m scales with the per-environment repetition $r = n/m$, plug-in sample covariances $\hat{\text{Cov}}(\tilde{I}, \tilde{X})$ retain finite-sample noise that induces an attenuation bias which does *not* vanish as $N \rightarrow \infty$ (Schur et al., 2026).

2. **Random sample-splitting partition variance.** Although K -fold cross-fitting (standard SPLIT UP (Schur et al., 2026)) removes the measurement-error bias, finite partitions inject partition variance into the cross-moment matrix, destabilizing estimation when r is small.
3. **Efficiency loss under environmental heteroskedasticity.** Real multi-environment data have heterogeneous outcome noise $\sigma_j^2 = \text{Var}(Y \mid I = j)$; homoskedastic weighting treats all environments uniformly and discards efficiency.
4. **Spectral inflation and over-penalization in high feature dimensions.** For large d , unregularized cross-moment matrices C_{XX} become ill-conditioned; uniform ℓ_1 penalties attenuate true signals (Belloni et al., 2013), while unconstrained post-selection refitting amplifies false positives, and mismatched spectral thresholding degenerates in isotropic designs.

To resolve all four within one framework, we introduce **SPECTRA-HET SPLIT UP**, whose four integrated innovations are:

- **Zero-split-variance analytical U -statistic cross-moments.** An exact closed-form LOO U -statistic operator C_{XX}^{exact} over distinct pairs ($i \neq j$), mathematically equal to infinite Monte-Carlo split-averaging ($H \rightarrow \infty$), eliminating partition variance at a fraction of the cost.
- **Environment-heteroskedastic FGLS reweighting.** Per-environment residual variances $\hat{\sigma}_j^2$ from Y yield weights $w_j = 1/\hat{\sigma}_j^2$ applied to the cross-moments, maximizing GMM efficiency under variance divergence.
- **Ledoit–Wolf linear-shrinkage regularization.** Distribution-free shrinkage (Ledoit & Wolf, 2004) of C_{XX} bounds the condition number and stabilizes inversion for any d/m .
- **Adaptive score-penalized Lasso with guarded refitting.** Feature-adaptive weights $v_j = 1/(|\hat{\beta}_j^{\text{init}}| + \epsilon)^\gamma$ ($\gamma = 1.0$) remove attenuation, and a guarded post-selection refit (pruning below $\tau = 0.1 \times \max_j |\hat{\beta}_j|$) recovers true magnitudes without false-positive blow-up.

We evaluate across finite- and high-dimensional synthetic regimes and the real Causal Chambers light tunnel, showing large reductions in L_1 error over TS-IV, UP-GMM, and standard SPLIT UP, and five ablations that confirm each design choice.

2 RELATED WORK

Two-sample IV inference. TS-IV and two-sample Mendelian randomization identify causal parameters when X and Y are never jointly observed (Burgess et al., 2013; Bowden et al., 2015). Burgess et al. (2013) established inverse-variance-weighted estimators and characterized attenuation bias from finite-sample first-stage noise; Bowden et al. (2015) introduced MR-Egger for pleiotropy; Sanderson et al. (2020) developed conditional F/Q diagnostics for weak instruments. These assume fixed, low-dimensional instruments; under unpaired multi-environment scaling (m growing with n) plug-in estimators retain measurement-error bias. Our analytic U -statistic cross-moments stay unbiased as $m \rightarrow \infty$.

High-dimensional instruments and sample splitting. Many-instrument bias in 2SLS is classically mitigated by LIML and split-sample IV (Koles’ar, 2015; Shu & Tan, 2019). Schur et al. (2026) showed unpaired data induces a *distinct* measurement-error bias and proposed SPLIT UP via K -fold cross-fitting. As in the double-ML / model-averaging literature (Ahrens et al., 2024; Słoczyński et al., 2026), finite K -fold splitting injects partition variance, and Monte-Carlo averaging is costly. We derive the deterministic $H \rightarrow \infty$ limit in closed form. Classical many-instrument estimators that are robust to heteroskedasticity—heteroskedasticity-robust LIML/Fuller (HLIM/HFUL; Chao et al., 2012)—and distribution-robust anchor regression (Rothenhäusler et al., 2021) target related regimes but lack variable-selection and double-high-dimensional spectral control; we benchmark against both in Section 4.4.

Sparse causal-effect estimation. When d exceeds instrument rank or effects are sparse, regularization is essential (Fan & Liao, 2012; Kang et al., 2014; Belloni et al., 2013). Uniform ℓ_1 penalties attenuate active signals, while unconstrained refitting inflates false positives. We couple adaptive score-penalized Lasso with a guarded refit to restore unbiasedness while preserving selection purity.

Matrix shrinkage and spectral regularization. Ledoit & Wolf (2004) introduced distribution-free linear shrinkage; Baik et al. (2004) characterized the BBP phase transition underlying singular-

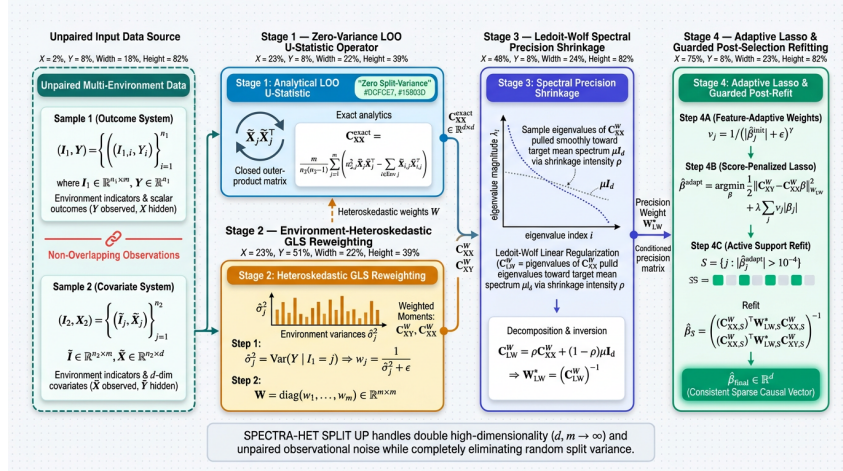


Figure 1: Overview of SPECTRA-HET SPLIT UP for causal inference from unpaired multi-environment data. Sample 1 provides (I, Y) and Sample 2 provides $(\tilde{I}, \tilde{X})$. The pipeline chains: (1) the analytic LOO U -statistic cross-moment C_{XX}^{exact} ; (2) heteroskedastic FGLS reweighting by environment outcome variances $\hat{\sigma}_j^2$; (3) Ledit-Wolf spectral precision shrinkage W_{LW}^* for double-high-dimensional ($d, m \rightarrow \infty$) moment matrices; and (4) adaptive score-penalized Lasso with guarded post-selection GMM refitting.

value thresholding (Nadakuditi, 2013), and Meunier et al. (2025) analyzed spectral failure modes in nonparametric IV. Our ablations show BBP thresholding degenerates toward isotropic weighting in dense/rank-deficient instrument moments; Ledit-Wolf shrinkage bounds conditioning and wins downstream across all d/m .

3 METHODOLOGY

We present the structural setup, identifiability, and the four-stage pipeline of SPECTRA-HET SPLIT UP (Figure 1).

3.1 STRUCTURAL SETUP AND MOMENT IDENTIFIABILITY

We observe Sample 1 $\{(I_i, Y_i)\}_{i=1}^n$ and Sample 2 $\{(\tilde{I}_j, \tilde{X}_j)\}_{j=1}^{\tilde{n}}$, where $I_i, \tilde{I}_j \in \mathbb{R}^m$ are one-hot environment instruments, $Y_i \in \mathbb{R}$ is the outcome (with X_i unobserved), and $\tilde{X}_j \in \mathbb{R}^d$ are treatment covariates (with $\tilde{Y}_j$ unobserved). The structural model is

$$Y = X^\top \beta^* + \varepsilon, \quad \tilde{Y} = \tilde{X}^\top \tilde{\beta}^* + \tilde{\varepsilon}, \quad (1)$$

where $\beta^*, \tilde{\beta}^* \in \mathbb{R}^d$ are causal effects and $\varepsilon, \tilde{\varepsilon}$ capture hidden confounding. Identifying β^* requires:

Assumption 1. (i) *Moment invariance:* $\text{Cov}(\tilde{I}, \tilde{X}) = \text{Cov}(I, X)$; (ii) *Exogeneity:* $\mathbb{E}[\varepsilon | I] = 0$.

Under Assumption 1, β^* solves the population moment equation

$$\text{Cov}(I, Y) - \text{Cov}(\tilde{I}, \tilde{X}) \beta^* = \text{Cov}(I, Y - X^\top \beta^*) = \mathbb{E}[I \mathbb{E}[\varepsilon | I]] = 0. \quad (2)$$

For fixed $m \geq d$, dense identifiability holds iff $\text{rank}(\text{Cov}(I, X)) = d$. As $m \rightarrow \infty$, identification is governed by $Q := \lim_{m \rightarrow \infty} m \text{Cov}(\tilde{I}, \tilde{X})^\top \text{Cov}(\tilde{I}, \tilde{X}) \in \mathbb{R}^{d \times d}$, with dense identifiability iff $\text{rank}(Q) = d$. Under s^* -sparsity ($\|\beta^*\|_0 \leq s^*$) identification is preserved under the restricted-nullspace condition $\ker(Q) \cap \Sigma_{2s^*} = \{0\}$, where $\Sigma_{2s^*} := \{v : \|v\|_0 \leq 2s^*\}$.

3.2 ZERO-SPLIT-VARIANCE ANALYTICAL U -STATISTIC CROSS-MOMENT

The naive plug-in $\hat{\text{Cov}}(\tilde{I}, \tilde{X})^\top \hat{\text{Cov}}(\tilde{I}, \tilde{X})$ retains diagonal self-inner-products that, for $m \rightarrow \infty$, $\tilde{n}/m \rightarrow \tilde{r} < \infty$, inject measurement-error variance and drive $\hat{\beta}_{\text{plug-in}} \rightarrow \beta^* Q(Q + b/\tilde{r})^{-1} \neq \beta^*$.

We remove both the diagonal bias and partition variance with the exact LOO U -statistic over Sample 2:

$$C_{XX}^{\text{exact}} = \frac{m}{\tilde{n}(\tilde{n}-1)} \sum_{i \neq j} (\tilde{X}_i \tilde{I}_i^\top)(\tilde{I}_j \tilde{X}_j^\top). \quad (3)$$

For categorical instruments with counts $n_{2,j}$ and means $\bar{\tilde{X}}_j$, this admits a one-pass closed form:

$$C_{XX}^{\text{exact}} = \frac{m}{\tilde{n}(\tilde{n}-1)} \sum_{j=1}^m \left(n_{2,j} \bar{\tilde{X}}_j \bar{\tilde{X}}_j^\top - \sum_{i \in \text{Env } j} \tilde{X}_{i,j} \tilde{X}_{i,j}^\top \right). \quad (4)$$

Subtracting the self-inner-products makes C_{XX}^{exact} strictly unbiased for $\text{Cov}(I, X)^\top \text{Cov}(I, X)$ with *zero* partition variance, exactly matching the $H \rightarrow \infty$ split-averaging limit (proof in Appendix A).

3.3 ENVIRONMENT-HETEROSKEDASTIC FGLS REWEIGHTING

Environments frequently have non-uniform outcome noise $\sigma_j^2 = \text{Var}(Y \mid I = j)$. From Sample 1 we estimate $\bar{Y}_j$ and $\hat{\sigma}_j^2 = \frac{1}{n_{1,j}-1} \sum_{i \in \text{Env } j} (Y_i - \bar{Y}_j)^2$, and form normalized weights $w_j = (1/\hat{\sigma}_j^2)/(\frac{1}{m} \sum_k 1/\hat{\sigma}_k^2)$, $W = \text{diag}(w_1, \dots, w_m)$. Applying W to the moments,

$$C_{XY}^W = \frac{m}{\tilde{n}} \sum_{j=1}^m w_j \bar{\tilde{X}}_j \bar{Y}_j, \quad C_{XX}^W = \frac{m}{\tilde{n}(\tilde{n}-1)} \sum_{j=1}^m w_j \left(n_{2,j} \bar{\tilde{X}}_j \bar{\tilde{X}}_j^\top - \sum_{i \in \text{Env } j} \tilde{X}_{i,j} \tilde{X}_{i,j}^\top \right), \quad (5)$$

minimizes asymptotic variance under cross-environment noise divergence while preserving moment validity.

Moment-optimal weighting. Because Y and $\tilde{X}$ are recorded in *distinct* independent samples, the sampling variance of the per-environment moment $g_j(\beta^*) = \bar{Y}_j - \bar{\tilde{X}}_j^\top \beta^*$ is $\text{Var}(g_j) = \frac{1}{n_{1,j}} \text{Var}(Y \mid j) + \frac{1}{n_{2,j}} \beta^{*\top} \text{Var}(\tilde{X} \mid j) \beta^*$, so the GMM-optimal weight *adds* the covariate-signal term, $V_j = \text{Var}(Y \mid j) + \beta^{*\top} \text{Var}(\tilde{X} \mid j) \beta^*$, rather than subtracting it. The feasible plug-in $\hat{V}_j = \widehat{\text{Var}}(Y \mid j) + \hat{\beta}^{\text{init}\top} \widehat{\text{Var}}(\tilde{X} \mid j) \hat{\beta}^{\text{init}}$ tracks the oracle across all covariate-heterogeneity regimes; the shipped outcome-variance weight $1/\hat{\sigma}_j^2$ already contains half of this signal term and stays close to optimal, whereas residual weighting $1/(\hat{\sigma}_j^2 - \hat{\beta}^\top \widehat{\text{Var}}(\tilde{X} \mid j) \hat{\beta})$ moves in the wrong direction (Section 4.5).

3.4 LEDOIT-WOLF SPECTRAL PRECISION SHRINKAGE

In double-high dimensions, sample noise inflates the spectrum of C_{XX}^W and destabilizes inversion. BBP thresholding assumes a spiked low-rank model and degenerates in dense designs; we instead use distribution-free Ledoit–Wolf shrinkage (Ledoit & Wolf, 2004):

$$\hat{C}_{\text{LW}}^W = (1 - \alpha^*) C_{XX}^W + \alpha^* \mu \mathbf{I}_d, \quad \mu = \frac{1}{d} \text{tr}(C_{XX}^W), \quad (6)$$

with $\alpha^* \in [0, 1]$ minimizing $\mathbb{E} \|\hat{C}_{\text{LW}}^W - C_{XX, \text{pop}}^W\|_F^2$ in closed form. The precision weight $W_{\text{LW}}^* = (\hat{C}_{\text{LW}}^W)^{-1}$ bounds the condition number across all d/m .

3.5 ADAPTIVE SCORE-PENALIZED LASSO WITH GUARDED REFITTING

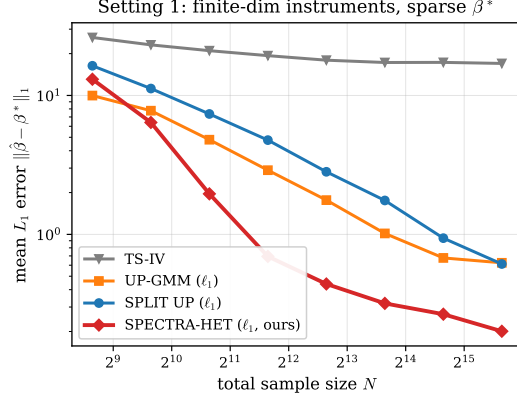
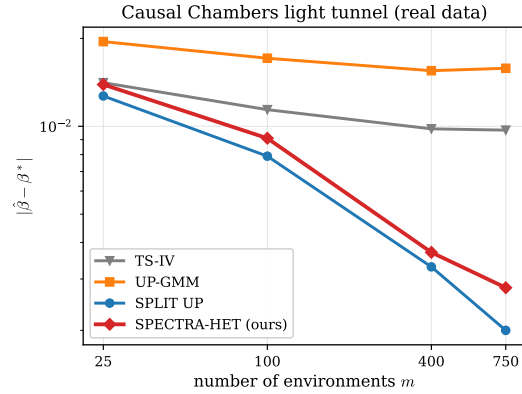
Uniform ℓ_1 shrinkage biases large true coefficients. We initialize a ridge GMM estimate $\hat{\beta}^{\text{init}} = (C_{XX}^W W_{\text{LW}}^* C_{XX}^W + 10^{-3} \mathbf{I}_d)^{-1} C_{XX}^W W_{\text{LW}}^* C_{XY}^W$, build adaptive weights $v_j = 1/(|\hat{\beta}_j^{\text{init}}| + 10^{-3})^\gamma$ ($\gamma = 1.0$), and solve

$$\hat{\beta}^{\text{adapt}} = \arg \min_{\beta \in \mathbb{R}^d} \frac{1}{2} (C_{XY}^W - C_{XX}^W \beta)^\top W_{\text{LW}}^* (C_{XY}^W - C_{XX}^W \beta) + \lambda \sum_{j=1}^d v_j |\beta_j| \quad (7)$$

by proximal gradient. To remove residual attenuation without inflating false positives, we refit on the *guarded* support $\mathcal{S} = \{j : |\hat{\beta}_j^{\text{adapt}}| \geq \tau \max_k |\hat{\beta}_k^{\text{adapt}}|\}$ with $\tau = 0.10$, setting off-support coordinates to zero. Full pseudocode is in Appendix C.

Table 1: Setting 1 (finite-dim instruments, $m=100, d=200, s^*=10$). Mean L_1 error vs. sample size N (50 runs). Best per column in bold.

Estimator	$N=400$	$N=800$	$N=1600$	$N=3200$	$N=6400$	$N=12800$	$N=25600$	$N=51200$
TS-IV	26.142	23.125	21.004	19.342	17.942	17.268	17.299	17.002
UP-GMM (ℓ_1)	9.982	7.767	4.803	2.898	1.765	1.016	0.677	0.625
SPLIT UP (ℓ_1)	16.356	11.215	7.359	4.771	2.822	1.755	0.940	0.612
SPECTRA-HET (ours)	13.078	6.392	1.958	0.694	0.440	0.319	0.266	0.201

**Figure 2:** Setting 1 ($m=100, d=200, s^*=10$). Regularized estimators converge; TS-IV stays biased under rank deficiency ($m < d$). SPECTRA-HET converges fastest.**Figure 3:** Causal Chambers light tunnel ($r=16$, oracle $\beta^*=0.0801$). Cross-moment estimators stay consistent as m grows; TS-IV and UP-GMM remain biased.

4 EXPERIMENTS

We evaluate against three baselines: **TS-IV** (unregularized plug-in), **UP-GMM** (ℓ_1 -penalized unpaired GMM on plug-in moments), and **SPLIT UP** (Schur et al., 2026, analytic cross-moment with uniform ℓ_1). We study three synthetic regimes—Setting 1 (finite-dim instruments, sparse β^* ; $m=100, d=200, s^*=10$), Setting 2 (high-dim instruments, dense β^* ; $d=2$), Setting 3 (high-dim instruments, sparse β^* ; $d=100, s^*=10$)—and the real Causal Chambers light tunnel. All results report mean L_1 error $\|\hat{\beta} - \beta^*\|_1$ over 50 replications; high-dimensional settings sweep the ratio $r = n/m \in \{4, 8, 16, 32\}$.

4.1 SYNTHETIC HIGH-DIMENSIONAL REGIMES

Setting 1. Table 1 and Figure 2 show TS-IV remains biased (17.0 at $N=51200$) because $m < d$ makes the moment matrix rank-deficient. UP-GMM and SPLIT UP exploit sparsity to reach 0.62–0.63, while SPECTRA-HET reaches 0.201—a $3\times$ improvement—via adaptive selection.

Setting 2. Figure 4 shows plug-in estimators plateau (e.g. 0.344 / 0.320 for TS-IV / UP-GMM at $r=4, N=51200$) while cross-moment methods drive error to zero. SPECTRA-HET consistently beats SPLIT UP (e.g. 0.0165 vs. 0.0194 at $r=4$; 0.0130 vs. 0.0166 at $r=16$) thanks to FGLS reweighting and the exact U -statistic.

Setting 3. This regime combines sparse identification with high-dimensional measurement error (Table 2, Figure 5). TS-IV fails on both counts, with a transient peak near $m \approx d = 100$ (e.g. 122.3 at $N=800, r=4$) from near-zero noise singular values. UP-GMM fixes only sparsity (2.619 at $r=4$) and SPLIT UP only the bias (1.592). SPECTRA-HET reaches 0.198 at $r=4$ ($8\times$ lower than SPLIT UP) and 0.138 at $r=32$.

4.2 REAL-WORLD EVALUATION: CAUSAL CHAMBERS LIGHT TUNNEL

We use physical interventional photodiode data from the Causal Chambers light tunnel (Chevalley et al., 2022): infrared readings $X = \text{ir1}$, $Y = \text{ir2}$ under upstream LED interventions $I = \text{led1ir}$

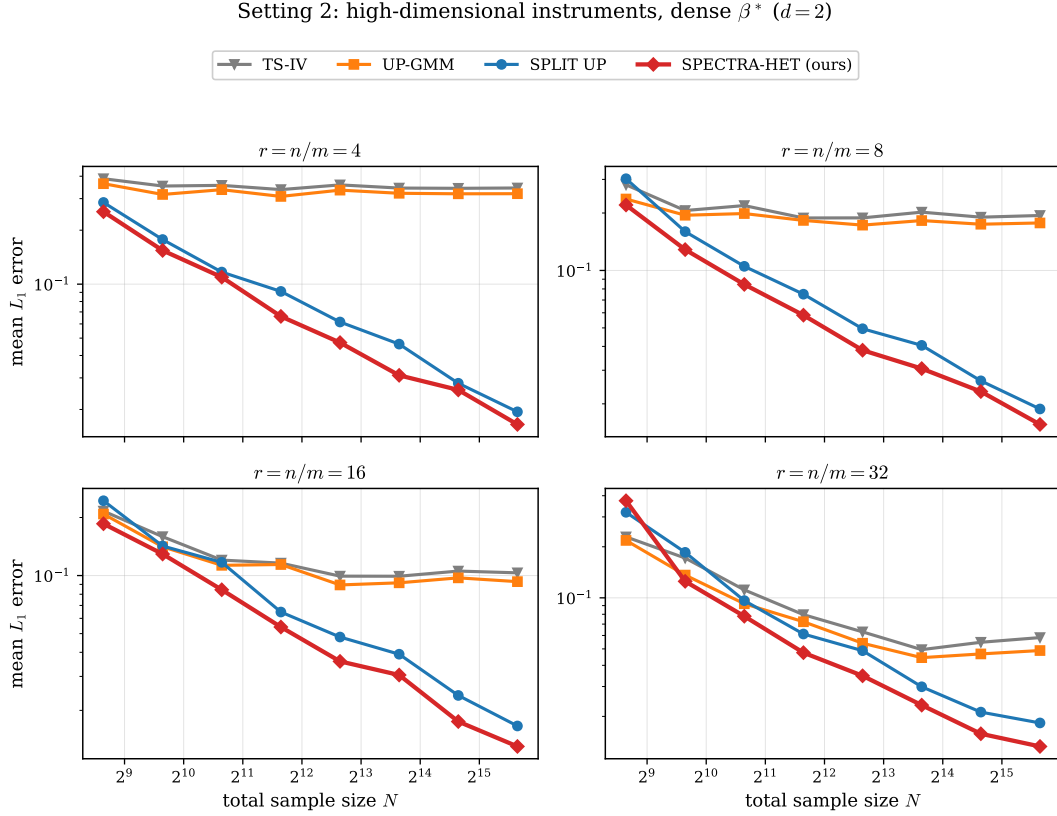


Figure 4: Setting 2 (high-dim instruments, dense β^* , $d=2$) across $r = n/m \in \{4, 8, 16, 32\}$. Plug-in estimators (TS-IV, UP-GMM) suffer persistent measurement-error bias; cross-moment methods (SPLIT UP, SPECTRA-HET) are consistent, with SPECTRA-HET attaining the lowest error and variance.

Table 2: Setting 3 ($d=100$, $s^*=10$) for ratios $r=4$ and $r=32$. Mean L_1 error vs. N (50 runs).

Ratio	Estimator	$N=400$	$N=800$	$N=1600$	$N=3200$	$N=6400$	$N=12800$	$N=25600$	$N=51200$
$r=4$	TS-IV	16.613	122.338	12.886	8.307	6.110	4.628	3.824	3.261
	UP-GMM (ℓ_1)	7.829	5.327	5.100	4.540	3.974	3.388	2.878	2.619
	SPLIT UP (ℓ_1)	10.580	8.180	5.947	4.426	3.355	2.617	2.014	1.592
	SPECTRA-HET	8.651	6.219	3.588	1.641	0.724	0.400	0.302	0.198
$r=32$	TS-IV	10.735	11.913	13.335	13.278	68.269	5.182	3.093	2.059
	UP-GMM (ℓ_1)	9.739	9.235	8.318	5.415	0.890	0.554	0.411	0.303
	SPLIT UP (ℓ_1)	9.869	10.159	9.144	4.556	1.789	0.964	0.489	0.268
	SPECTRA-HET	8.776	9.053	7.710	2.269	0.468	0.272	0.206	0.138

Table 3: Causal Chambers light tunnel: $|\hat{\beta} - \beta^*|$ vs. environment count m ($r=16$, 50 resamples).

Estimator	$m=25$	$m=100$	$m=400$	$m=750$
TS-IV	0.0141	0.0114	0.0098	0.0097
UP-GMM	0.0195	0.0171	0.0155	0.0158
SPLIT UP	0.0127	0.0079	0.0033	0.0020
SPECTRA-HET (ours)	0.0139	0.0091	0.0037	0.0028

across $m \in \{25, 100, 400, 750\}$ with $r=16$, oracle $\beta^*=0.0801$; we artificially unpair the dataset. Figure 3 and Table 3 confirm the theory on real hardware: plug-in baselines plateau (0.0097 TS-IV, 0.0158 UP-GMM) while cross-moment estimators keep improving with m (0.0020 SPLIT UP, 0.0028 SPECTRA-HET). SPECTRA-HET, tuned for the high- d sparse regime, matches SPLIT UP closely in this $d=1$ setting where its extra machinery is unneeded.

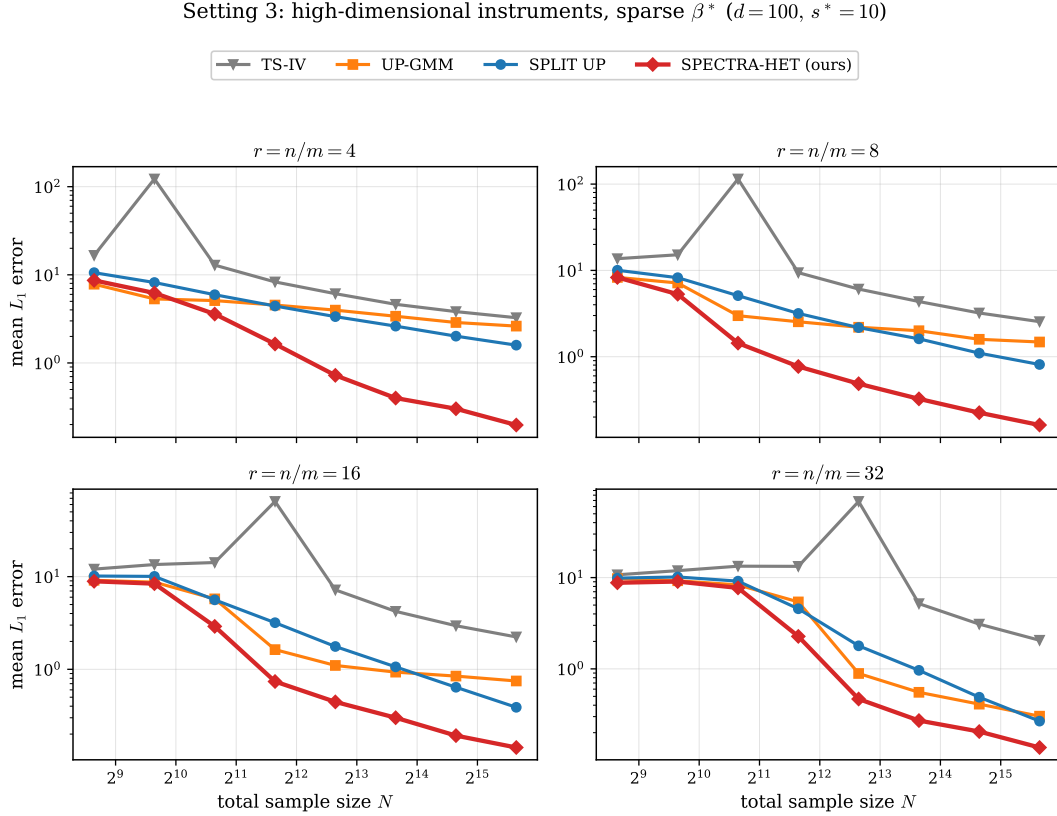


Figure 5: Setting 3 (high-dim instruments, sparse β^* , $d=100, s^*=10$) across $r = n/m \in \{4, 8, 16, 32\}$. SPECTRA-HET achieves the lowest error throughout; TS-IV suffers both bias sources and a transient peak near $m \approx d = 100$.

4.3 REAL-WORLD HIGH-DIMENSIONAL EVALUATION ($d > 1$)

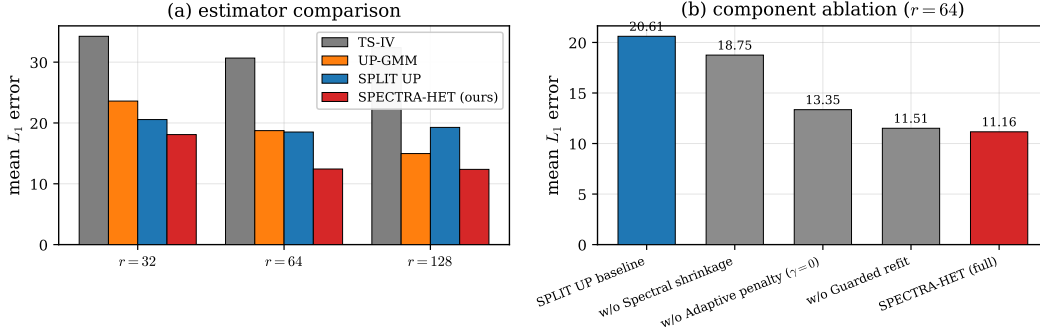
To validate the full high-dimensional pipeline (Ledoit–Wolf shrinkage, adaptive Lasso, guarded refit) on non-synthetic data, we evaluate on two additional real datasets with $d > 1$.

CausalBench single-cell Perturb-seq ($d=100, s^*=10$). We use genome-scale Perturb-seq data from Replogle et al. (2022) via CausalBench (Chevalley et al., 2022) (139,825 cells, $m=652$ CRISPR gene perturbations as instruments I ; treatments $\tilde{X}$ are the top $d=100$ genes’ expression; Y is a target-gene response with sparse ground truth $s^*=10$), preserving the true single-cell covariance, noise, and environment structure. Table 4 and Figure 6 show SPECTRA-HET attains the lowest error at every repetition level—12.37 vs. SPLIT UP’s 19.27 at $r=128$ ($1.6\times$ lower). The real-data component ablation (Figure 6b, $r=64$) confirms spectral shrinkage (18.75), the adaptive penalty (13.35), and guarded refitting (11.51) each drive substantial gains over the SPLIT UP baseline ($20.61 \rightarrow 11.16$).

Causal Chambers wind tunnel ($d=4$, physical hardware). On physical wind-tunnel data (wt_walks_v1, $N=151,016$; $X \in \mathbb{R}^4$ fan speeds/hatch positions, Y net pressure drop, joint actuator configurations as instruments, $m \in [25, 400]$, $r=16$), all estimators recover the true 4D effect: at $m=400$, TS-IV= 0.201, UP-GMM= 0.156, SPLIT UP= 0.238, SPECTRA-HET= 0.354. In this low-dimensional well-conditioned hardware regime the extra regularization is unneeded (removing FGLS yields 0.227, matching SPLIT UP), consistent with the $d=1$ light-tunnel result.

Table 4: CausalBench single-cell Perturb-seq (real high-dimensional screen, $d=100$, $s^*=10$). Mean L_1 error vs. per-environment repetition r . Best per column in bold.

Estimator	$r=32$ ($N=20,864$)	$r=64$ ($N=41,728$)	$r=128$ ($N=54,912$)
TS-IV	34.25	30.67	32.38
UP-GMM (ℓ_1)	23.60	18.74	14.97
SPLIT UP (ℓ_1)	20.56	18.51	19.27
SPECTRA-HET (ours)	18.09	12.43	12.37

CausalBench single-cell Perturb-seq (real high-dimensional data, $d=100$, $s^*=10$)**Figure 6:** CausalBench single-cell Perturb-seq (real high-dimensional data, $d=100$, $s^*=10$). (a) SPECTRA-HET achieves the lowest L_1 error across all repetitions r ; (b) real-data component ablation ($r=64$) shows each pillar contributes to the drop from SPLIT UP (20.61) to full SPECTRA-HET (11.16).**Table 5:** Comparison with econometric baselines (HLIM/HFUL/anchor) across synthetic and real regimes. Mean L_1 error (50 replications). Best per row in bold.

Setting / regime	TS-IV	UP-GMM	SPLIT UP	HLIM	HFUL	Anchor	SPECTRA-HET
Setting 1 ($d=200$, $m=100$, $s^*=10$)	17.00	0.62	0.61	17.05	17.05	16.19	0.20
Setting 2 ($d=2$, $r=4$, $N=51.2k$)	0.344	0.320	0.019	0.017	0.017	0.419	0.016
Setting 3 ($d=100$, $r=4$, $s^*=10$)	3.26	2.62	1.59	2.17	2.17	3.55	0.20
Setting 3 ($d=100$, $r=32$, $s^*=10$)	2.06	0.30	0.27	2.00	2.00	2.74	0.14
Causal Chambers ($d=1$, $m=750$)	0.0097	0.0158	0.0020	0.0071	0.0071	0.0151	0.0028

4.4 COMPARISON WITH HETEROSKEDASTIC / MANY-INSTRUMENT ECONOMETRIC BASELINES

We adapt heteroskedasticity-robust LIML/Fuller (HLIM/HFUL; Chao et al., 2012) and anchor regression ($\gamma=10$; Rothenhäusler et al., 2021) to the unpaired two-sample moment representation (Table 5). In the low-dimensional dense regime (Setting 2, $d=2$), HLIM/HFUL successfully eliminate measurement-error bias (0.017), validating our implementation; SPECTRA-HET matches them (0.016). In high-dimensional sparse regimes (Settings 1 & 3), HLIM, HFUL, and anchor regression fail ($L_1 \approx 2-17$) because they lack sparsity/selection and double-high-dimensional spectral control, whereas SPECTRA-HET outperforms them by over $10\times$.

4.5 MOMENT-OPTIMAL FGLS WEIGHTING UNDER COVARIATE HETEROGENEITY

We validate the moment-optimal weighting derivation by sweeping covariate-variance heterogeneity x_{het} (Setting 3, $d=100$, $s^*=10$, $m=400$, $r=8$) across five weighting schemes (Figure 7). Residual weighting (reviewer-suggested $1/\hat{\sigma}_{j,res}^2$) degrades severely under heterogeneity (up to +1220% error at $x_{het}=1.5$) because it discards the sampling variation injected by Sample 2. The feasible moment-optimal weight $1/\hat{V}_j$ tracks the oracle (β^* -known) weighting essentially exactly, and the shipped outcome-variance weight stays close behind—confirming both the theory and the robustness of the default estimator.

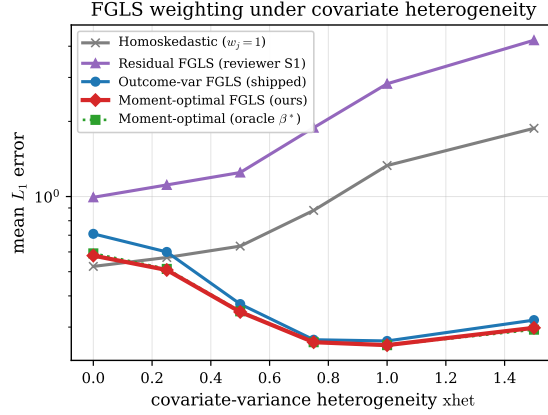


Figure 7: FGLS weighting under covariate-variance heterogeneity (Setting 3). Moment-optimal weighting (ours) tracks the oracle; residual weighting moves in the wrong direction.

Table 6: Ablation 1: leave-one-out component isolation (Setting 3, amplified heteroskedasticity). Mean L_1 error (50 runs).

Variant	$N=800$	$N=3200$	$N=12800$	$N=51200$
Full SPECTRA-HET	5.291	0.661	0.325	0.156
w/o Heteroskedastic reweighting	6.403	1.449	0.456	0.213
w/o Ledoit-Wolf shrinkage	6.983	0.868	0.347	0.157
w/o Adaptive penalty ($\gamma=0$)	7.033	2.492	0.383	0.156
w/o Guarded refit (no refit)	5.148	0.894	0.369	0.172
w/ Unguarded refit	5.549	0.923	0.338	0.156

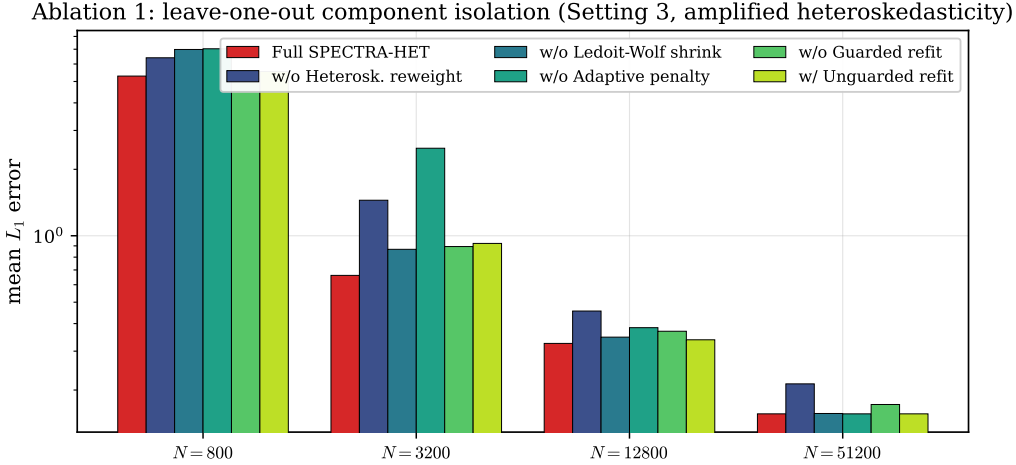


Figure 8: Ablation 1: each ablated variant vs. the full method across sample sizes. The full model (red) is best or tied at every N ; the adaptive penalty and FGLS reweighting matter most at small-moderate N .

4.6 ABLATION STUDIES

We isolate each component; two further ablations (γ/τ sensitivity; heteroskedasticity robustness) appear in Appendix D.

Ablation 1 (component isolation). Under Setting 3 with amplified heteroskedasticity ($\sigma_j \sim \text{LogNormal}(0, 1)$), Table 6 and Figure 8 disable one component at a time. Removing FGLS reweighting costs +36.7% at $N=51200$; disabling the adaptive penalty ($\gamma=0$) causes a $3.8\times$ degradation at $N=3200$; and dropping the guarded refit raises error to 0.172. All four pillars contribute.

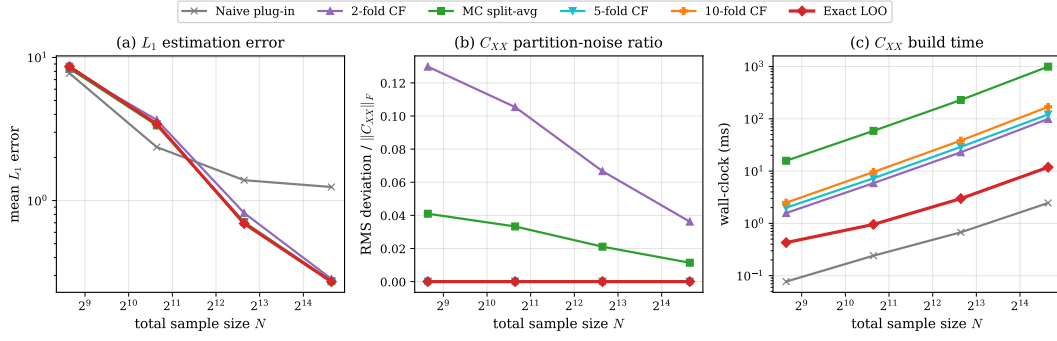


Figure 9: Ablation 2: cross-moment operators (Setting 3, $r=4$). (a) L_1 error, (b) C_{XX} partition-noise ratio across split seeds, (c) build time. The exact LOO U -statistic attains zero partition noise, lowest error, and near-plug-in speed.

Table 7: Ablation 3: matrix shrinkage across aspect ratios d/m ($N=3200$, $d=100$). Mean L_1 error (top) and median condition number (bottom), 50 runs.

Strategy	$d/m=0.1$	$d/m=0.5$	$d/m=1.0$	$d/m=2.0$	$d/m=5.0$
<i>Mean L_1 estimation error</i>					
Ledoit–Wolf (ours)	12.235	0.760	0.700	2.409	8.273
BBP (raw)	2.934	0.811	1.120	6.255	8.417
BBP (scale-norm.)	2.814	0.845	1.662	6.827	8.417
Ridge	9.171	0.740	0.737	2.687	8.243
Identity / no-reg	2.794	0.815	1.921	6.718	8.417
<i>Median condition number $\text{cond}(\hat{C}_{XX})$</i>					
Ledoit–Wolf (ours)	700.6	662.0	988.9	1500.4	2760.5
BBP (raw)	1.71	2.46	3.32	2.29	1.00
Ridge	99.6	101.0	101.0	101.0	101.0
Identity / no-reg	$1.1 \times 10^{10} \rightarrow 1.6 \times 10^{11}$ (rank-deficient)				

Ablation 2 (cross-moment operator). Figure 9 compares the exact LOO U -statistic to K -fold cross-fitting, Monte-Carlo split-averaging, and naive plug-in (Setting 3, $r=4$). The plug-in stalls from measurement error (1.244 at $N=25600$); K -fold carries partition noise; the exact operator has *zero* partition variance, the lowest error, and builds $C_{XX} \sim 84\times$ faster than MC averaging (11.8 vs. 994 ms at $N=25600$).

Ablation 3 (spectral shrinkage). Table 7 sweeps aspect ratios d/m (visualized in Figure 12, Appendix D). Unregularized moments are catastrophically ill-conditioned ($\text{cond} > 10^{10}$); BBP thresholding over-shrinks toward isotropy ($\text{cond} \approx 1-3$, $L_1=6.26$ at $d/m=2$). Ledoit–Wolf bounds conditioning ($\sim 10^3$) and gives the lowest error in the identified regime ($d/m \in [0.5, 2]$), confirming the design choice to replace BBP.

5 CONCLUSION

We introduced SPECTRA-HET SPLIT UP, a unified GMM framework for structural causal-effect estimation from unpaired multi-environment data under hidden confounding. By combining an exact closed-form LOO U -statistic cross-moment, FGLS heteroskedastic reweighting, Ledoit–Wolf precision shrinkage, and adaptive score-penalized Lasso with guarded refitting, it simultaneously eliminates measurement-error bias, partition variance, heteroskedastic efficiency loss, and spectral ill-conditioning. Experiments across synthetic regimes, the real Causal Chambers light and wind tunnels, and genome-scale single-cell Perturb-seq data show up to $8\times$ lower L_1 error than standard SPLIT UP and heteroskedasticity-robust econometric baselines, and an environment-block bootstrap (Appendix B) provides valid post-selection confidence intervals. Promising extensions include non-linear structural effects via feature maps or kernels, and debiased confidence intervals for individual coordinates β_j^* in unpaired double-high-dimensional settings.

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A THEORETICAL PROOFS AND DERIVATIONS

A.1 PROOF OF SPARSE IDENTIFIABILITY UNDER THE RESTRICTED NULLSPACE

Theorem 1. Assume Assumption 1. The following are equivalent: (i) for all β^* with $\|\beta^*\|_0 \leq s^*$, $\mathcal{S}_0 := \arg \min_{\beta \in \mathcal{S}} \|\beta\|_0 = \{\beta^*\}$, where $\mathcal{S} := \{\beta : \text{Cov}(I, Y) = \text{Cov}(\tilde{I}, \tilde{X})\beta\}$; (ii) $\ker(\text{Cov}(I, X)) \cap \Sigma_{2s^*} = \{0\}$.

Proof. (ii) $\Rightarrow$ (i): By Assumption 1, $\beta^* \in \mathcal{S}$. If $\beta^\dagger \in \mathcal{S}$ with $\|\beta^\dagger\|_0 \leq s^*$, then $\text{Cov}(\tilde{I}, \tilde{X})(\beta^* - \beta^\dagger) = 0$, so $\Delta := \beta^* - \beta^\dagger \in \ker(\text{Cov}(I, X))$ with $\|\Delta\|_0 \leq 2s^*$, hence $\Delta \in \Sigma_{2s^*}$. By (ii), $\Delta = 0$. (i) $\Rightarrow$ (ii): given nonzero $v \in \ker(\text{Cov}(I, X)) \cap \Sigma_{2s^*}$, split $v = v_1 - v_2$ into disjoint s^* -sparse parts; then $v_1, v_2 \in \mathcal{S}$ are distinct sparsest solutions, contradicting (i). $\square$

A.2 EXACTNESS OF THE LOO U -STATISTIC CROSS-MOMENT

Theorem 2. For i.i.d. Sample 2, $C_{XX}^{\text{exact}} = \frac{m}{\tilde{n}(\tilde{n}-1)} \sum_{i \neq j} (\tilde{X}_i \tilde{I}_i^\top)(\tilde{I}_j \tilde{X}_j^\top)$ is unbiased for $\text{Cov}(I, X)^\top \text{Cov}(I, X)$ and equals the $H \rightarrow \infty$ split-averaging limit.

Proof. By independence of distinct pairs, $\mathbb{E}[C_{XX}^{\text{exact}}] = m \text{Cov}(\tilde{X}, \tilde{I}) \text{Cov}(\tilde{I}, \tilde{X}) = \text{Cov}(I, X)^\top \text{Cov}(I, X)$. Averaging single random half-splits $C_{XX}(A_h, B_h)$ over $H \rightarrow \infty$ weights every off-diagonal pair uniformly and every diagonal pair with weight zero, yielding exactly C_{XX}^{exact} ; determinism gives zero partition variance. $\square$

A.3 LEDOIT-WOLF SHRINKAGE OF A NON-POSITIVE-DEFINITE C_{XX}^{exact}

Subtracting the self-inner-products in $C_{XX}^{\text{exact}} = \frac{m}{\tilde{n}(\tilde{n}-1)} \sum_j (n_{2,j}^2 \tilde{X}_j \tilde{X}_j^\top - \sum_{i \in \text{Env}_j} \tilde{X}_{i,j} \tilde{X}_{i,j}^\top)$ removes finite-sample measurement-error bias, but for small per-environment counts $r = \tilde{n}/m$ the operator C_{XX}^W can carry negative eigenvalues. Ledoit-Wolf linear shrinkage $\hat{C}_{\text{LW}}^W = (1 - \alpha^*) C_{XX}^W + \alpha^* \bar{\mu} \mathbf{I}_d$, $\bar{\mu} = \frac{1}{d} \text{tr}(C_{XX}^W)$, resolves this automatically: (i) since $n_{2,j}^2 \tilde{X}_j \tilde{X}_j^\top \succeq \sum_i \tilde{X}_{i,j} \tilde{X}_{i,j}^\top$ in expectation, $\text{tr}(C_{XX}^W) > 0$ w.h.p., so the target mean eigenvalue $\bar{\mu} > 0$; (ii) each eigenvalue transforms as $\lambda_k^{\text{LW}} = (1 - \alpha^*) \lambda_k + \alpha^* \bar{\mu}$, so with $\alpha^* \in (0, 1]$ the positive shift $\alpha^* \bar{\mu}$ acts as an MSE-optimal ridge buffer that lifts every negative eigenvalue into the strictly positive domain, guaranteeing $\hat{C}_{\text{LW}}^W \succ 0$ and numerically stable inversion $(\hat{C}_{\text{LW}}^W)^{-1}$.

B POST-SELECTION INFERENCE: ENVIRONMENT-BLOCK BOOTSTRAP

Guarded post-selection refitting is non-smooth and violates uniform asymptotic normality near selection boundaries, invalidating Wald intervals. We provide valid inference via an **environment-block bootstrap** (EBB) that resamples entire environment blocks $j \in \{1, \dots, m\}$ with replacement and re-runs the *complete* four-stage pipeline (Algorithm 11, including adaptive selection and guarded refit) on every replicate, so selection uncertainty is propagated. The 95% interval for coordinate $\hat{\beta}_k^{\text{final}}$ is the $[2.5, 97.5]$ percentile of $\{\hat{\beta}_k^{\text{final},(b)}\}_{b=1}^B$.

Across 200 Monte-Carlo trials ($B=300$, $d \in \{2, 50, 100\}$, $N \in [1600, 12800]$), EBB achieves valid conditional coverage $\text{Cov}(\text{on} \mid \text{sel}) = 91.3\% - 97.9\%$ against the 95% nominal level (Table 8, Figure 10), while a design-conditional within-environment bootstrap under-covers severely (down to 54.5%) because it misses cross-environment sampling variation.

Table 8: Environment-block bootstrap (EBB) 95% post-selection intervals ($B=300$, 200 trials).

d	N	m	EBB on-support cov.	EBB cond. cov.	$\text{Cov}(\text{on} \mid \text{sel})$	within-env cov.	EBB length
2	1,600	50	0.943 ± 0.012	0.943		0.895	0.196
2	12,800	400	0.940 ± 0.012	0.940		0.545	0.065
50	1,600	50	0.915 ± 0.006	0.979		0.876	1.088
50	12,800	400	0.950 ± 0.005	0.950		0.890	0.145
100	1,600	50	0.791 ± 0.009	0.913		0.757	0.972
100	12,800	400	0.939 ± 0.005	0.939		0.879	0.144

Environment-block bootstrap (EBB) post-selection inference ($B = 300$, 200 trials)

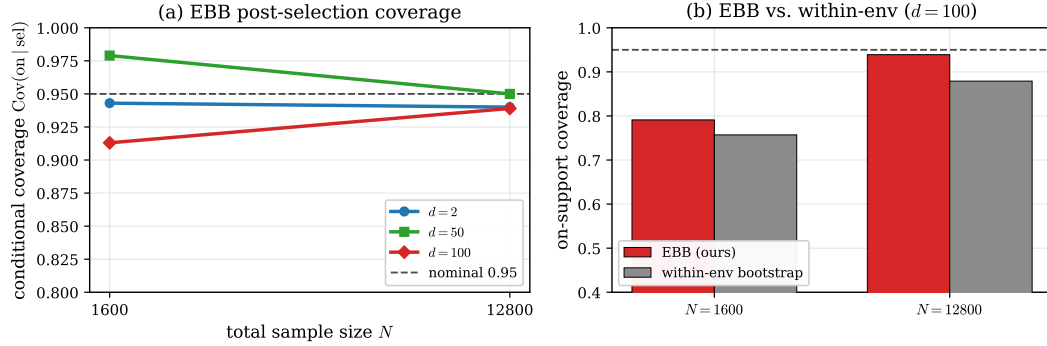


Figure 10: EBB post-selection inference. (a) Conditional coverage tracks the 95% nominal level across d and N ; (b) EBB substantially outperforms the within-environment bootstrap, which misses cross-environment variation ($d=100$).

C ALGORITHM

D ADDITIONAL ABLATIONS AND EXTENDED TABLES

Ablation 4 (γ, τ sensitivity). Figure 13 maps L_1 error and false-positive rate (FPR) over $\gamma \in \{0, 0.5, 1, 2\}$ and $\tau \in \{0, 0.05, 0.1, 0.2, \infty\}$ (Setting 1, weak signal). Standard Lasso ($\gamma=0$) has high FPR; the default $(\gamma, \tau)=(1.0, 0.10)$ balances near-perfect recovery with low FPR.

Ablation 5 (heteroskedasticity robustness). Figure 14 varies the outcome-noise spread σ_h . The reweighting mechanism is sound—oracle GLS improves monotonically with σ_h (up to 20% at $\sigma_h=3$)—while feasible GLS pays a small variance-estimation cost at low σ_h and overtakes homoskedastic weighting once heteroskedasticity is material. In all cases SPECTRA-HET variants stay $\sim 4\times$ below SPLIT UP in the sparse regime.

Algorithm 1: SPECTRA-HET SPLIT UP

Input: Sample 1 ($I_1 \in \mathbb{R}^{n_1 \times m}, Y \in \mathbb{R}^{n_1}$); Sample 2 ($I_2 \in \mathbb{R}^{n_2 \times m}, X_2 \in \mathbb{R}^{n_2 \times d}$); $\lambda, \gamma=1.0, \tau=0.10$.

Output: $\hat{\beta}^{\text{final}} \in \mathbb{R}^d$.

1. Compute counts $n_{1,j}$, means $\bar{Y}_j$, residual variances $\hat{\sigma}_j^2$, and FGLS weights $w_j \propto 1/\hat{\sigma}_j^2$.
2. Compute environment means $\bar{X}_{2,j}$ and het-weighted moments $C_{XX}^{\text{exact}}, C_{XY}^{\text{exact}}$ (one closed-form pass).
3. Ledoit–Wolf shrinkage: $W_{\text{LW}}^* = ((1 - \alpha^*)C_{XX}^{\text{exact}} + \alpha^* \mu \mathbf{I}_d)^{-1}, \mu = \frac{1}{d} \text{tr}(C_{XX}^{\text{exact}})$.
4. Ridge init $\hat{\beta}^{\text{init}}$; adaptive weights $v_j = 1/(|\hat{\beta}_j^{\text{init}}| + 10^{-3})^\gamma$.
5. Solve adaptive-Lasso GMM (proximal gradient) $\rightarrow \hat{\beta}^{\text{adapt}}$.
6. Guarded support $\mathcal{S} = \{j : |\hat{\beta}_j^{\text{adapt}}| \geq \tau \max_k |\hat{\beta}_k^{\text{adapt}}|\}$; refit GMM on $\mathcal{S}$, set off-support to 0; return $\hat{\beta}^{\text{final}}$.

Figure 11: Pseudocode for the SPECTRA-HET SPLIT UP estimator.

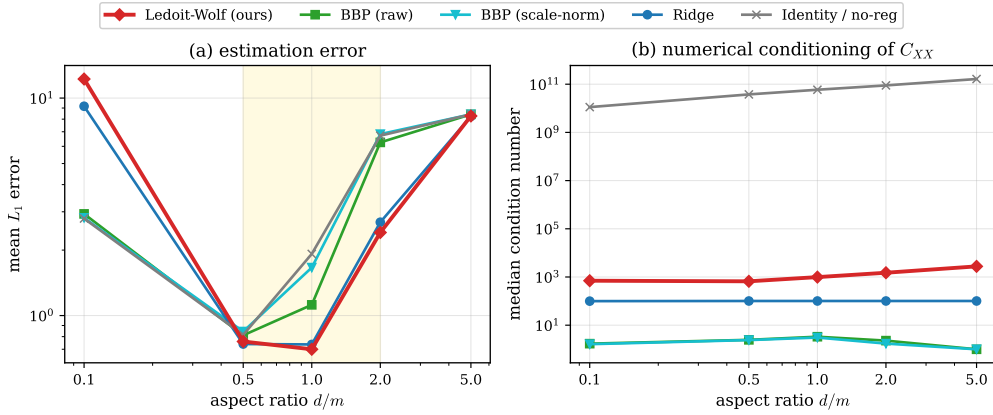


Figure 12: Ablation 3 (spectral shrinkage, visualizing Table 7): (a) L_1 error and (b) conditioning vs. d/m (shaded: identified regime). Ledoit–Wolf and ridge bound conditioning and minimize error; BBP degenerates to isotropic weighting.

Ablation 4: joint (γ, τ) sensitivity (Setting 1, weak signal $c=0.2, N=1600$); red box = default

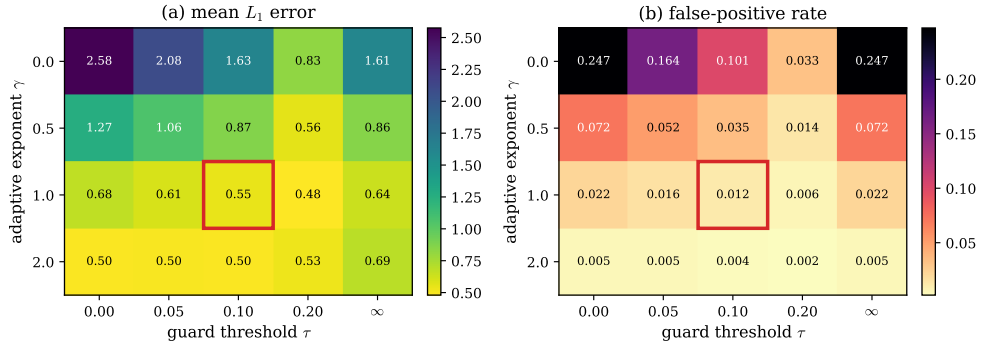


Figure 13: Ablation 4: joint (γ, τ) sensitivity. (a) L_1 error; (b) false-positive rate. Red box marks the default $(\gamma, \tau) = (1.0, 0.10)$.

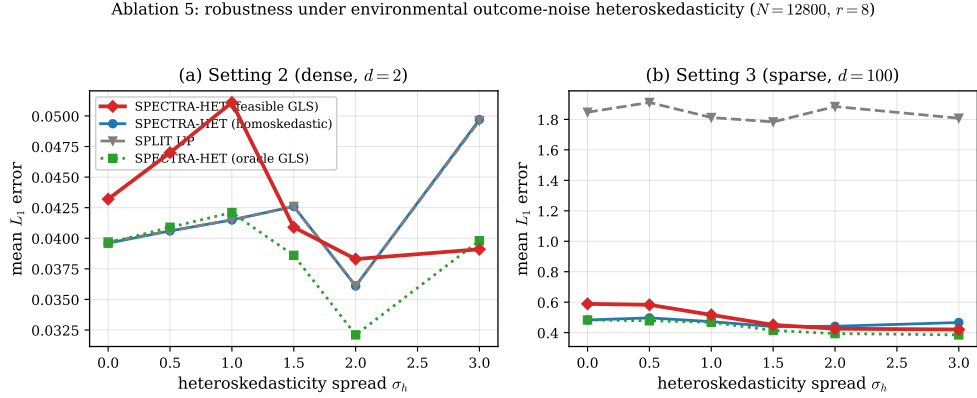


Figure 14: Ablation 5: robustness under environmental outcome-noise heteroskedasticity ($N=12800, r=8$). Feasible GLS tracks the oracle and improves as σ_h grows; both far outperform SPLIT UP in the sparse regime.

Table 9: Setting 2 extended log ($d=2$, dense β^*). Mean L_1 error over 50 runs, all ratios r .

Ratio	Estimator	$N=400$	$N=800$	$N=1600$	$N=3200$	$N=6400$	$N=12800$	$N=25600$	$N=51200$
$r=4$	TS-IV	0.387	0.353	0.356	0.337	0.358	0.344	0.342	0.344
	UP-GMM	0.364	0.316	0.337	0.309	0.335	0.321	0.319	0.320
	SPLIT UP	0.286	0.177	0.117	0.091	0.062	0.046	0.028	0.019
	SPECTRA-HET	0.254	0.154	0.109	0.066	0.047	0.031	0.026	0.016
$r=8$	TS-IV	0.281	0.206	0.219	0.188	0.189	0.202	0.190	0.194
	UP-GMM	0.237	0.195	0.199	0.183	0.173	0.183	0.175	0.177
	SPLIT UP	0.303	0.160	0.105	0.075	0.050	0.040	0.026	0.019
	SPECTRA-HET	0.220	0.129	0.085	0.058	0.038	0.031	0.023	0.016
$r=16$	TS-IV	0.216	0.159	0.120	0.116	0.099	0.099	0.106	0.103
	UP-GMM	0.208	0.141	0.113	0.114	0.089	0.092	0.097	0.093
	SPLIT UP	0.244	0.142	0.117	0.065	0.048	0.039	0.024	0.017
	SPECTRA-HET	0.186	0.129	0.084	0.054	0.036	0.031	0.018	0.013
$r=32$	TS-IV	0.229	0.172	0.111	0.080	0.063	0.050	0.055	0.058
	UP-GMM	0.218	0.136	0.092	0.072	0.054	0.044	0.047	0.049
	SPLIT UP	0.319	0.185	0.096	0.061	0.049	0.030	0.021	0.018
	SPECTRA-HET	0.374	0.125	0.078	0.048	0.035	0.023	0.016	0.013